

Project Requirements

- **Team Theme:**
 - Work in teams (max 5 students) based on a shared **biocad theme** (e.g., Construction Files, Plasmid Visualization).
 - Focus on **individual work** within the team.
- **Function Development:**
 - Have **at least one MCP capability** that solves a problem related to your team's theme.
 - Must be **well-documented** and address a clear task.
- **MCP Wrappers:**
 - Include **MCP wrappers** with metadata (inputs, outputs, description).
 - Provide **test prompts** to trigger your Function based on the wrapper
- **Pytest:**
 - Provide **pytest(s)** to verify the function's correctness.

https://docs.google.com/presentation/d/1nIPA1EjRWerT5MKIWu6kOyFIEFVbDJ4-3-EM2tGoGd0/edit?slide=id.g30e4ec0a91a_0_6#slide=id.g30e4ec0a91a_0_6

Sign up for demo day

- You should have made prompts and wrappers for the capabilities in your project
- You are presenting the big picture of your project, and how the LLM interacts with your wrappers and prompts
- It doesn't need to be a polished presentation. Think more group meeting than formal presentation.
- But it is part of your grade to show effort and understanding of the project.
- All should contribute to the presentation, including individually scoped efforts
- Estimate 1 or 2 slides max * number of people on team (3-5 min)
- Combine your slides with your presentation group

Big Picture

A user describes a CRISPR experiment (target gene, organism, cell type) and the system walks them through the full design process: from guide RNA selection to validation strategy.

Guide RNA Designer	Off-Target Analyzer	Construction File Builder	Delivery Strategy Advisor	Validation Planner
Takes a target gene/sequence, returns ranked candidate gRNAs with on-target scores	Takes a gRNA and organism, returns risk-scored list of potential off-target sites	Takes Cas variant + target + kit choice, outputs a formatted construction file	Takes cell type + Cas system, uses an internal LLM call to recommend and compare delivery methods (lipofection, RNP, viral, etc.)	Takes edit type, locus, and delivery method, recommends a sequencing/assay strategy and outputs a validation checklist

Guide RNA Designer

- Example MCP prompts:

- What is the best gRNA to use for this target gene:

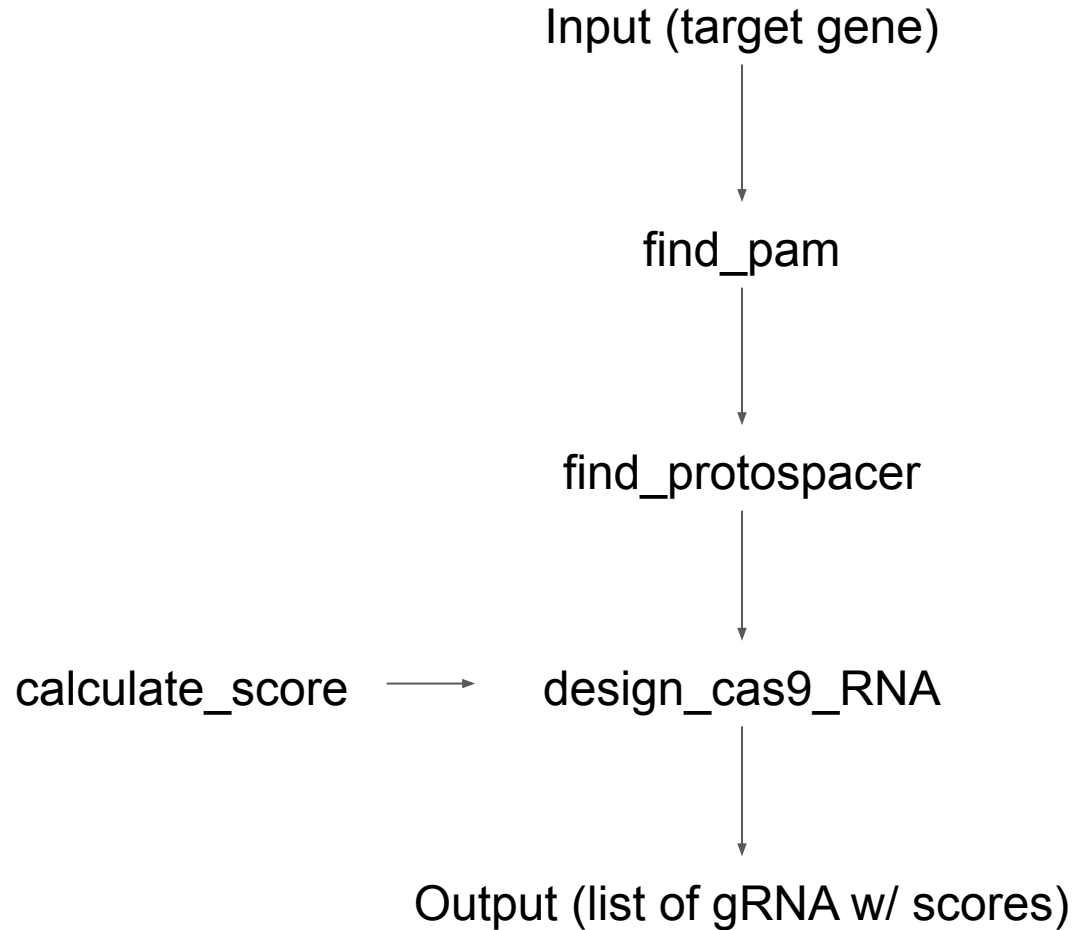
“CCCTAGATGCCTGGCTCAGAAACCTGCCAGTTTGCTGGCACGTTTTTTTCTTTTGCCTTT
AGTTCTCACGTTTGTGCTACTTGACAACGCTTCTTTAACCAAATATAATTGTTTACAGTACG
TTAGCAGTCTGGATGTACACG” in a CRISPR experiment?

- Input:

- “CCCTAGATGCCTGGCTCAGAAACCTGCCAGTTTGCTGGCACGTTTTTTTCTTTTGCCTTT
AGTTCTCACGTTTGTGCTACTTGACAACGCTTCTTTAACCAAATATAATTGTTTACAGTACG
TTAGCAGTCTGGATGTACACG”

- Output:

- “GUUUCUGAGCCAGGCAUCUAGUUUUAGAGCUAGAAAUAGCAAGUUAAAAUAAGGCUA
GUCCGUUAUCAACUUGAAAAAGUGGCACCGAGUCGGUGC” (Score: 100.0)
 - “UCAGAAACCUGCCAGUUUGCGUUUUAGAGCUAGAAAUAGCAAGUUAAAAUAAGGCUAG
UCCGUUAUCAACUUGAAAAAGUGGCACCGAGUCGGUGC” (Score: 100.0)



Off Target Analyzer

- Talk about

Construction File Builder

Example MCP prompts:

- Design a CRISPR construct for BRCA1 knockout using PX459.

Here is the input

```
{  
  "cas_variant": "SpCas9",  
  "target_gene": "BRCA1",  
  "organism": "human",  
  "guide_sequence": "GAGTCCGCAGTCTTAGAGCT",  
  "edit_type": "knockout",  
  "kit_choice": "PX459",  
  "off_target_risk_profile": {  
    "total_off_targets": 4,  
    "mismatch_distribution": {"0MM": 0, "1MM": 0, "2MM": 1, "3MM": 3}  
  }  
}
```

Construction File Builder

Delivery Strategy Advisor

Validation Planner